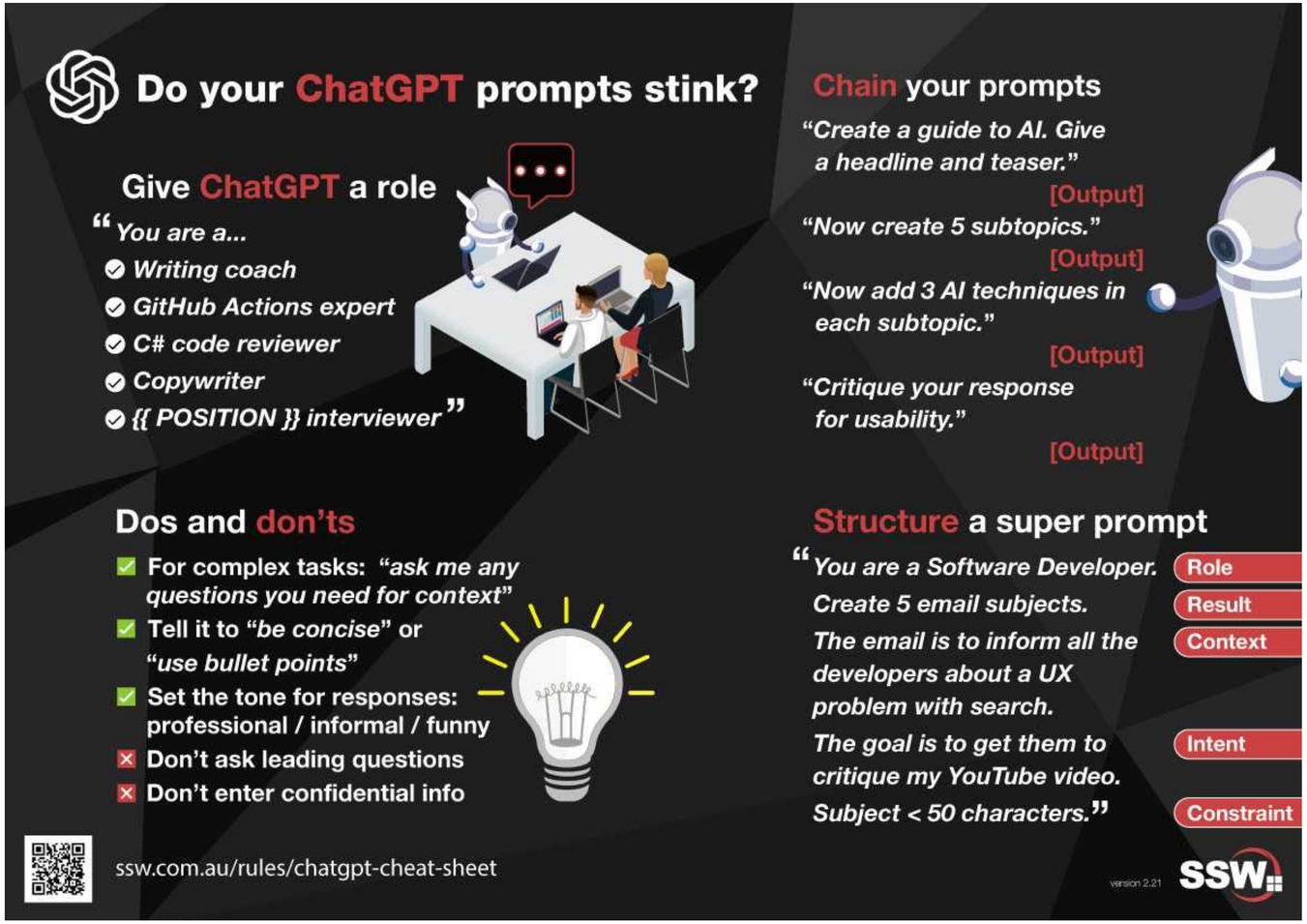
# Coding with Large Language Models (LLMs)

**May 28, 2025**
**IMET Workshop**

**Led by Burch Fisher**

# Prompt Engineering Primer

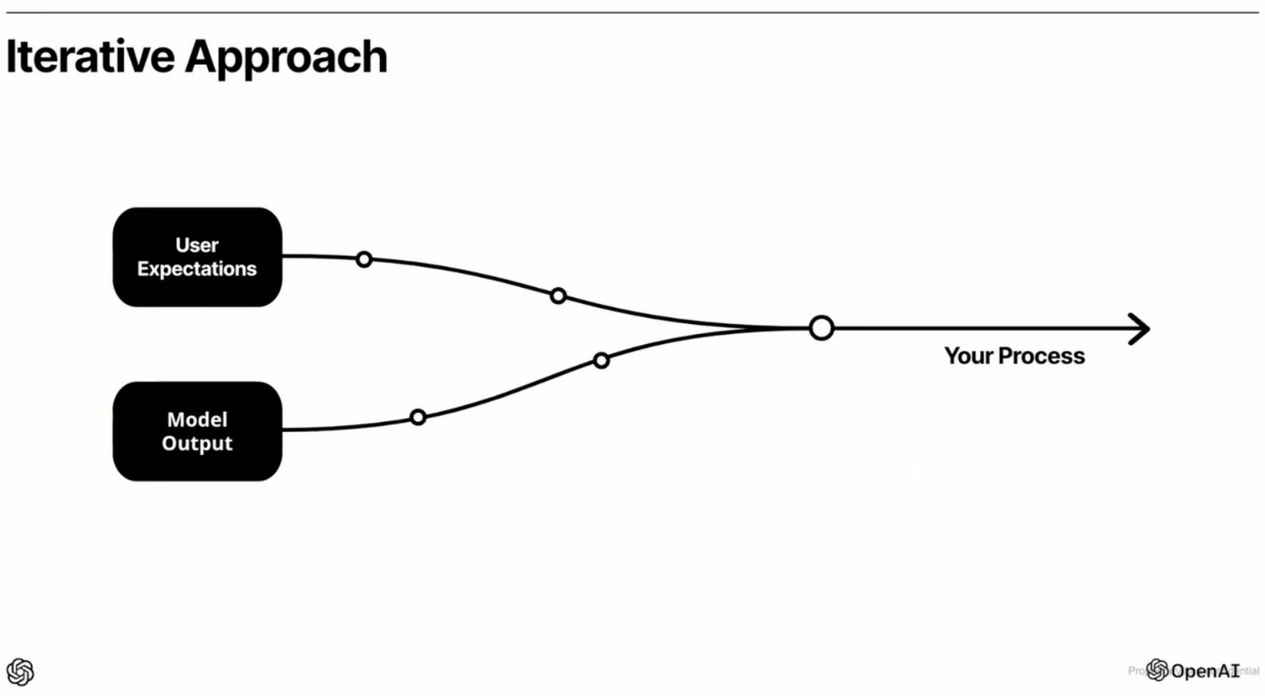
- **Prompt engineering** is the process of designing and optimizing the language input to effectively guide a LLM response.
- Effective prompt engineering requires **Context** and **Expectations** and can include a **Persona (Role)**
- In the same way you would communicate with a person, the more **context** and **detailed expectations** you provide the better the output will be from the LLM.
    - Think about walking into an ER to for triage




https://www.ssw.com.au/rules/static/342f93bb776e37da664f6ccc4e461764/be464/chatgpt-cheat-sheet-bathroom-door.jpg

# Prompt Engineering Primer

- **Prompt engineering** is an iterative process usually
- Can benefit from:
  - Delimiters can highlight logical breaks
  - Step-by-step instructions
  - One/few shot learning (e.g. give it examples of what you want)
  - Ask LLMs to help write effective prompts



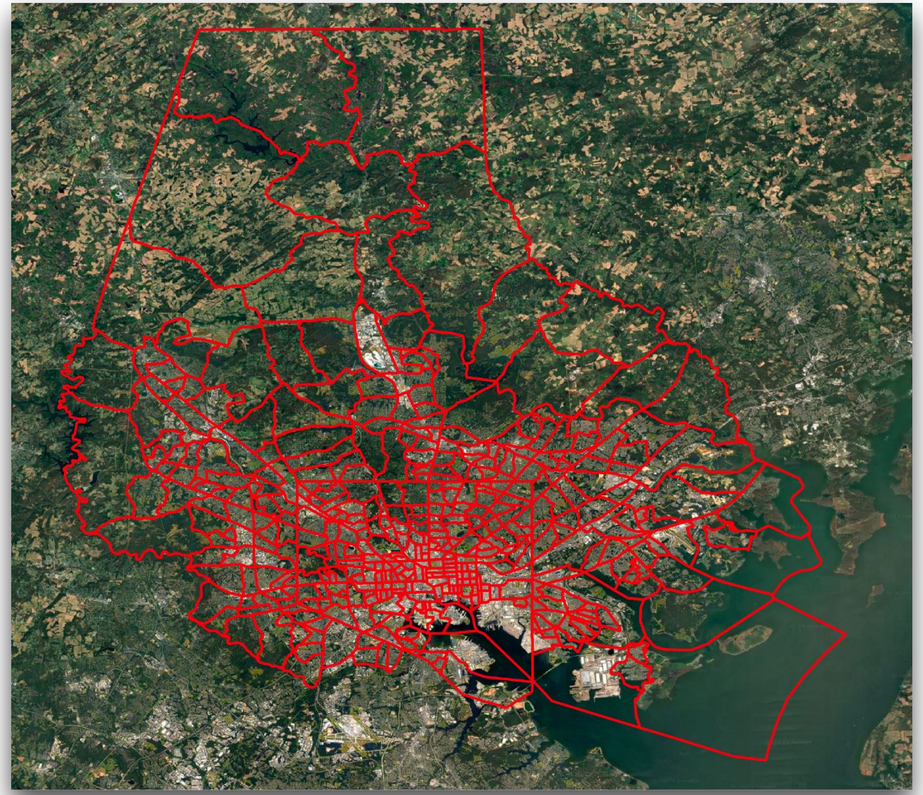

*https://academy.openai.com/public/videos/advanced-prompt-engineering-2025-02-13*

# Goals for the Session

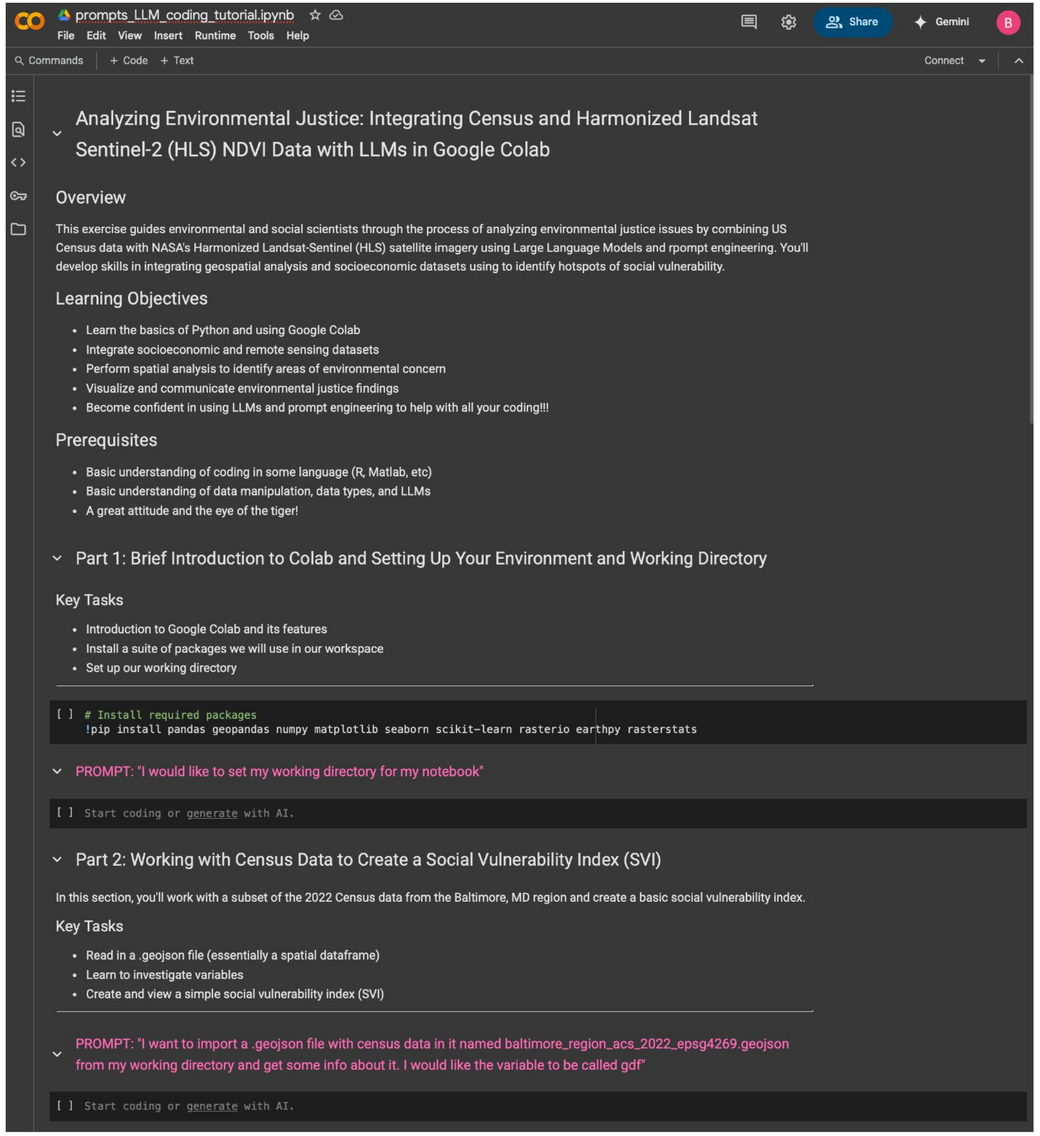
- Become comfortable working in **Google Colab** and building a workflow in **Python** even when you don't know Python!!
- Explore the **powerful capabilities** and **common pitfalls** of working with LLMs to code.
- **Have fun!!**

# Visual Roadmap for the Session

- **Part 1: Introduction to Google Colab environment and some very basic Python**



*Baltimore City and County Census Tracts we will use*



prompts_LLM_coding_tutorial.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text

Share Gemini Connect

## Analyzing Environmental Justice: Integrating Census and Harmonized Landsat Sentinel-2 (HLS) NDVI Data with LLMs in Google Colab

### Overview

This exercise guides environmental and social scientists through the process of analyzing environmental justice issues by combining US Census data with NASA's Harmonized Landsat-Sentinel (HLS) satellite imagery using Large Language Models and rpompt engineering. You'll develop skills in integrating geospatial analysis and socioeconomic datasets using to identify hotspots of social vulnerability.

### Learning Objectives

- Learn the basics of Python and using Google Colab
- Integrate socioeconomic and remote sensing datasets
- Perform spatial analysis to identify areas of environmental concern
- Visualize and communicate environmental justice findings
- Become confident in using LLMs and prompt engineering to help with all your coding!!!

### Prerequisites

- Basic understanding of coding in some language (R, Matlab, etc)
- Basic understanding of data manipulation, data types, and LLMs
- A great attitude and the eye of the tiger!

### Part 1: Brief Introduction to Colab and Setting Up Your Environment and Working Directory

Key Tasks

- Introduction to Google Colab and its features
- Install a suite of packages we will use in our workspace
- Set up our working directory

```
[ ] # Install required packages
    !pip install pandas geopandas numpy matplotlib seaborn scikit-learn rasterio earthpy rasterstats
```

PROMPT: "I would like to set my working directory for my notebook"

```
[ ] Start coding or generate with AI.
```

### Part 2: Working with Census Data to Create a Social Vulnerability Index (SVI)

In this section, you'll work with a subset of the 2022 Census data from the Baltimore, MD region and create a basic social vulnerability index.

Key Tasks

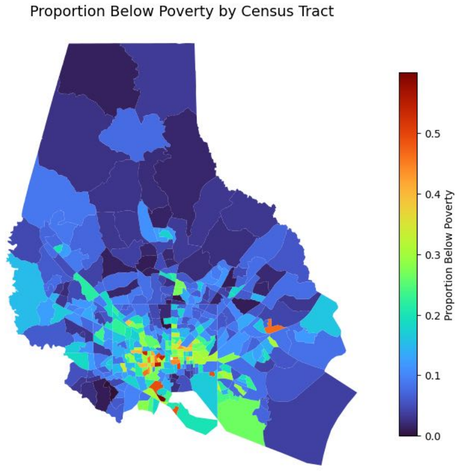
- Read in a .geojson file (essentially a spatial dataframe)
- Learn to investigate variables
- Create and view a simple social vulnerability index (SVI)

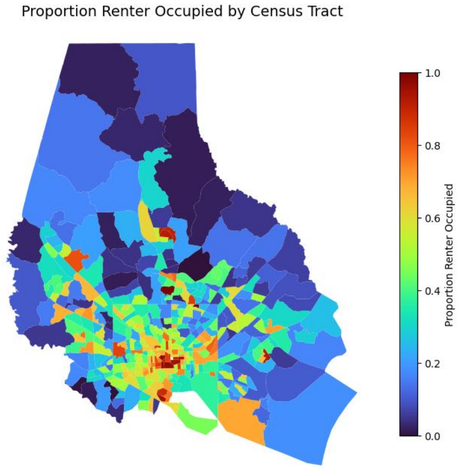
PROMPT: "I want to import a .geojson file with census data in it named baltimore_region_acs_2022_epsg4269.geojson from my working directory and get some info about it. I would like the variable to be called gdf"

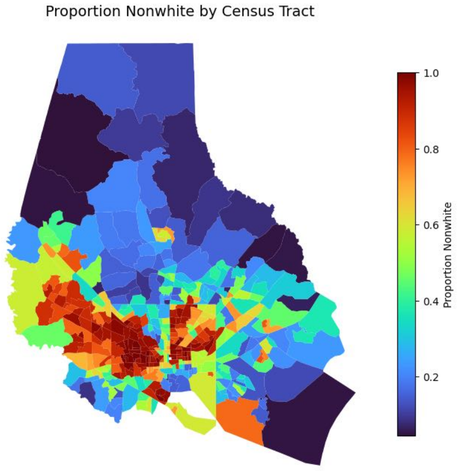
```
[ ] Start coding or generate with AI.
```

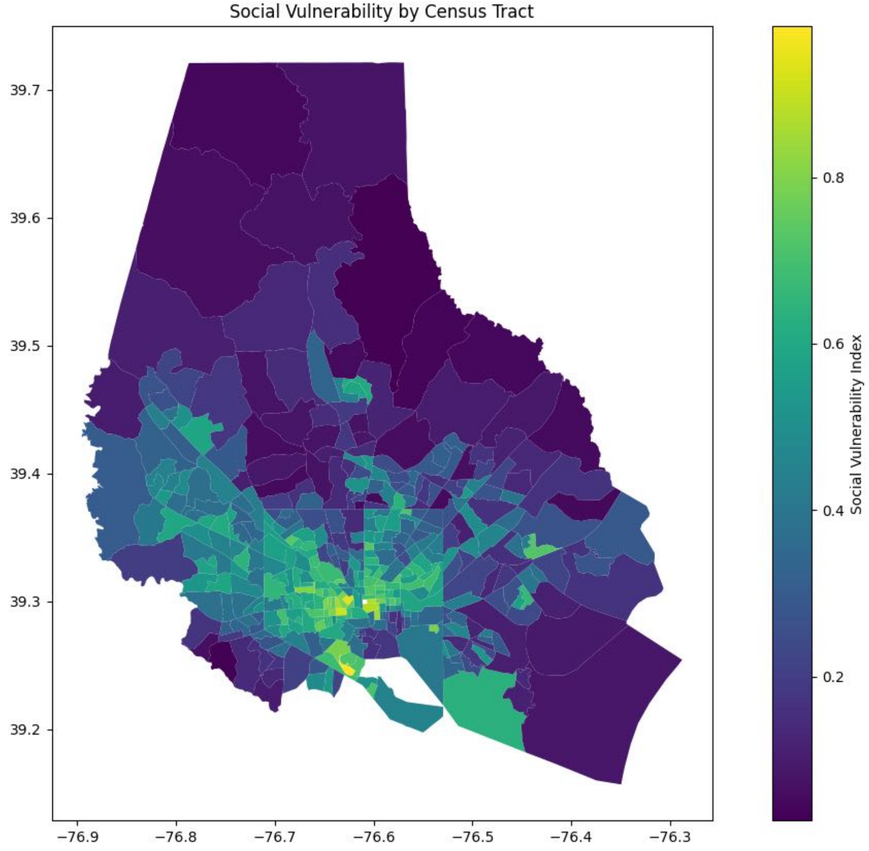
# Visual Roadmap for the Session

- **Part 2: Working with Census Data to Create a Social Vulnerability Index (SVI)**



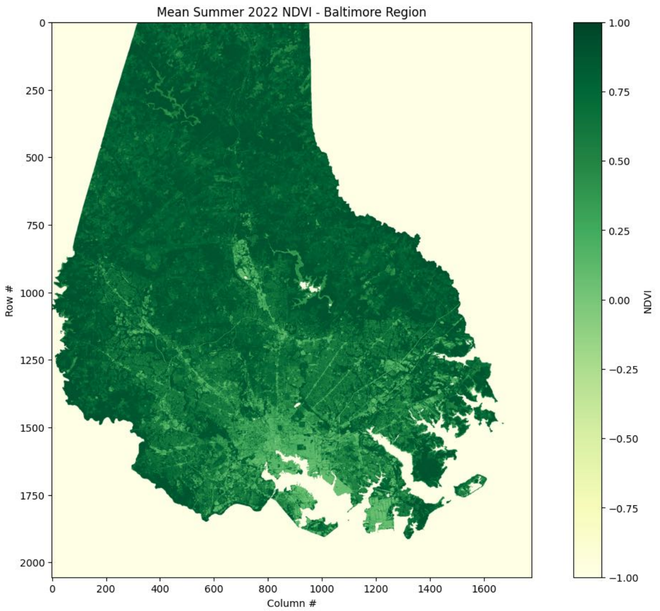



# Visual Roadmap for the Session

- **Part 3: Integrating Harmonized Landsat Sentinel-2 NDVI Data with Census Data to Illuminate Patterns of Social Vulnerability**

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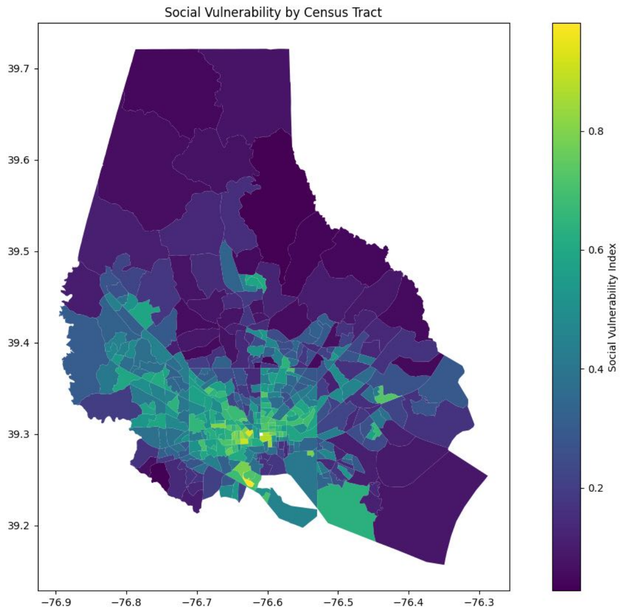


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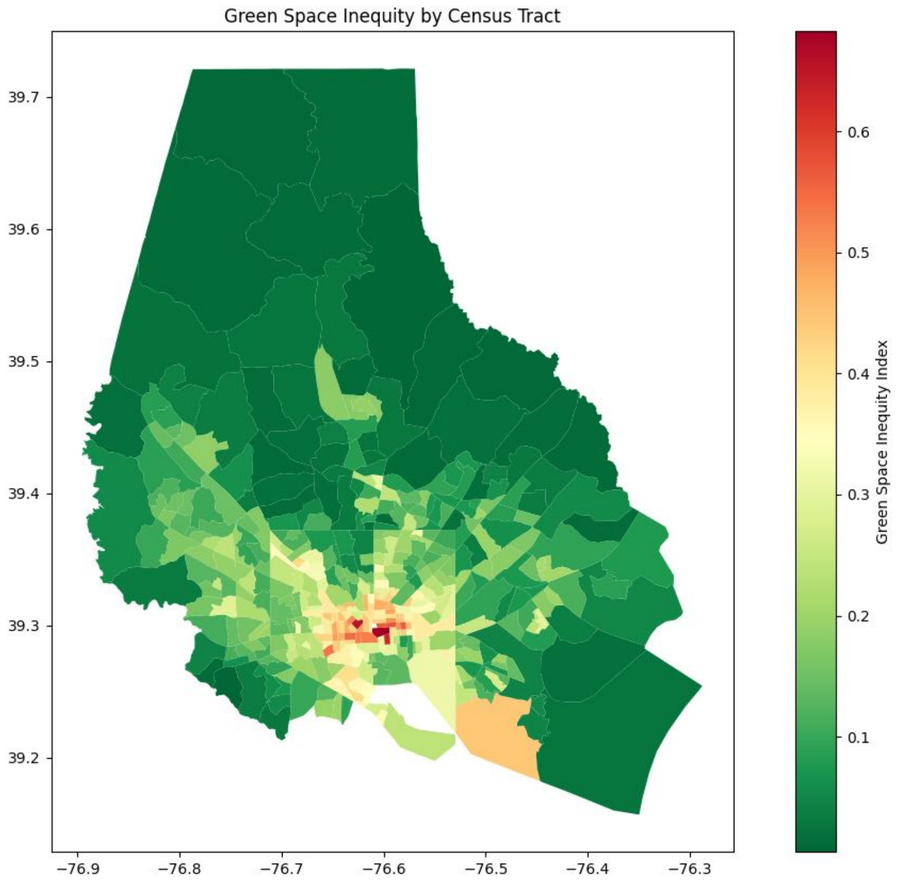

# Visual Roadmap for the Session

- **Part 4: Go Wild with LLM!!**

# Visual Roadmap for the Session

- **Part 5: Show and Tell: Creating a Dashboard for Environmental Justice Data Exploration Using an LLM**

**PROMPT to DALL-E:**

***Can you make an image that shows a cosmic large language model giving a gift to an environmental scientist?***